

Conditional Remedies and Statutory Design under Judicial Review

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Abstract

Many constitutional courts can preserve a statute while giving the legislature time to repair it rather than simply upholding or striking it down. We study how this remedial option changes ex ante legislative incentives in a model where the legislature chooses both statutory burden and repairability, and the court chooses among uphold, conditional relief, and strike. We show that adding conditional relief eliminates any global directional prediction for legislative aggressiveness. As repair productivity rises, the legislature's optimal statute can be aggressive, then nonaggressive, and then aggressive again. We also show that conditional relief creates an ex ante incentive to invest in repairability that is absent under binary review, under which repairability is optimally zero. The analysis identifies how middle-path remedies reshape statutory design before review occurs.

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1 Introduction

Judicial review in its canonical form is binary: a court either upholds a challenged statute or strikes it down. But many constitutional systems go further. The Canadian Supreme Court regularly issues suspended declarations of invalidity that keep a statute in force for a fixed period while the legislature rewrites it. South Africa’s Constitutional Court uses reading-in orders paired with legislative deferral. Germany’s Federal Constitutional Court can declare a law “incompatible” rather than void, directing the legislature to enact a compliant replacement. Indonesia’s Constitutional Court employs conditionally constitutional rulings that preserve a statute subject to a specified interpretive or legislative fix.¹

These remedies share a common structure: they preserve part or all of a statute while opening a window for legislative repair rather than immediately terminating the law. A legislature drafting under the shadow of such review faces a different calculus from one facing binary invalidation. Under binary review, the trade-off is between the policy gains from a more ambitious statute and the risk of outright loss. Under conditional review, the legislature also faces the possibility that an aggressive statute will survive in modified form, creating a second-chance channel that is absent from the binary benchmark.

I ask how that second-chance channel reshapes legislative incentives. I study a game between a legislature and a court in which expanding the judicial menu from uphold-or-strike to uphold, conditionally repair, or strike changes the legislature’s ex ante problem on two margins: the burden choice, which governs how aggressively the legislature drafts, and the repairability design, which governs how easy the statute is to fix if a court opens a repair window.

I establish two main results. The first, and more important, concerns the direction of discipline. A long tradition in the formal study of judicial review emphasizes the “shadow of review” as a disciplining device that shapes legislation before any court acts. Vanberg (1998) develops a model of abstract judicial review in which legislative bargaining is shaped by judicial constraints, and Rogers and Vanberg (2002, 2007) show that judicial review has an important passive effect on legislation and

¹On Canadian suspended declarations, see Roach (2004). On South Africa, Dixon (2007). On Germany, see Schlaich and Koriath (2004). On Indonesia, Butt and Murharjanti (2022).

can improve outcomes even under activist review. In these models, the court’s menu is effectively binary, and the discipline channel admits clean directional characterizations.

I show that once the remedial menu includes a conditional repair option, the sign of the discipline effect is no longer globally determinate. The core difficulty is that the conditional remedy changes the legislature’s continuation value in a way that depends on the interaction between repair productivity, the distribution of review states, and the geometry of the court’s remedy regions. To make this precise, I construct an explicit environment, inside the maintained class used throughout the paper, in which the legislature’s optimal statute is aggressive at a lower level of repair productivity, ceases to be aggressive at a higher level, and becomes aggressive again at a still higher value. The resulting aggressive-nonaggressive-aggressive pattern defeats any one-cutoff characterization in the repair-productivity parameter within the maintained environment. Because all comparisons in the counterexample are strict, the pattern is not a knife-edge artifact; it survives on small neighborhoods of the reference parameter values.

The counterexample isolates three mechanisms that are characteristic of conditional-remedy settings. First, truncation kinks arise when one review state’s repaired payoff is driven to zero while the other remains positive, creating a payoff discontinuity. Second, left-envelope re-expansion occurs when higher repair productivity improves the value of nonaggressive repaired statutes faster than aggressive ones, pulling the optimizer below the court’s strike threshold. Third, attainment failure on the aggressive side means the supremum over aggressive statutes may equal a boundary limit that is not attained, allowing the nonaggressive maximizer to dominate. These three mechanisms operate simultaneously in the counterexample, and their interaction is what generates the non-monotonicity. The counterexample is constructed within a specific nonlinear witness class (defined in Appendix A); while the mechanisms are suggestive of broader patterns, the formal result is established for this class.

The second result concerns statutory design. Conditional review creates an investment incentive that binary review does not. Under binary review, reparability design is behaviorally dormant: it matters only through the contingent-remand channel, which binary review excludes. The binary benchmark therefore optimally sets reparability at zero. Under conditional review, a well-designed

statute may survive where a poorly designed one would lapse, so the legislature has a positive marginal return to investing in repair-facilitating design features (severability scaffolding, fallback clauses, transition rules, delegated adjustment authority) whenever the conditional remedy is used with positive probability and the marginal benefit at zero exceeds the marginal design cost. In relating remedial structure to investment incentives, I am closest in spirit to Turner (2017), who studies how oversight regimes shape costly investments in policy capacity and shows that intermediate forms of oversight can induce either higher or lower investment depending on the regime. My contribution on this margin is to show that the conditional remedy creates a qualitative discontinuity: from zero investment under binary review to strictly positive investment under conditional review, rather than a continuous adjustment from one positive level to another.

I also connect to the law-and-economics literature on constitutional remedies. Fish (2016) maps the choice space of constitutional remedies. Leckey (2016) analyzes the costs of remedial discretion. Kavanagh (2015) examines the institutional consequences of weak-form review. I formalize the ex ante incentive consequences of one specific remedial expansion, from binary to conditional, rather than studying the full menu of remedial options. By focusing on a single well-defined menu expansion, I can identify the precise mechanisms through which conditional remedies reshape the legislature's optimal strategy within the model.

Related literature. My work sits at the intersection of the formal judicial politics literature and the comparative constitutional law literature on remedial design. On the formal side, Stephenson (2004) models judicial review as a disciplining device in which anticipation of court action constrains agency behavior; my model shares this “shadow of review” logic but expands the court's action set beyond the binary menu. Bueno de Mesquita and Stephenson (2002) study how informative precedent facilitates communication among judges, showing that opinion-writing serves a signaling function within the judiciary; my model abstracts from intrajudicial dynamics to isolate the remedial-menu channel, and I discuss in Section 8 how a strategic court might alter the results. Fox and Stephenson (2011) model judicial review as a response to political posturing, showing that review can improve policy even when the court is less informed; my model shares their focus on how

judicial institutions reshape legislative behavior but adds the conditional-remedy margin. Lax (2011) surveys how legal doctrine shapes judicial politics, emphasizing the strategic role of doctrinal rules as mechanism-design objects; my conditional remedy can be read as one such doctrinal expansion. Cameron (2000) studies presidential vetoes as a non-binary oversight structure, providing a template for analyzing how expanding the veto/review menu changes the bargaining game; my model applies analogous logic to judicial remedies. Shipan (2004) examines how institutional constraints shape statutory design, a theme closely connected to my investment result; the mechanism I identify is specific to conditional remedies rather than institutional constraints in general. On comparative constitutional remedies, Ginsburg and Versteeg (2014) study why countries adopt constitutional review, documenting cross-national variation in review institutions that provides context for the jurisdictional differences my model addresses, and Dixon and Ginsburg (2017) analyze the forms and limits of constitutions as political insurance, a framing that connects to the remedial-design question I study. Hogg and Bushell (1997) develop the concept of constitutional dialogue between courts and legislatures, which provides the institutional backdrop for the conditional-remedy channel I formalize. My model formalizes one element of that dialogue: the incentive consequences of giving courts a conditional repair option alongside uphold and strike.

The remainder of the paper proceeds as follows. Section 2 presents the model. Section 3 establishes the binary benchmark. Section 4 characterizes where the conditional remedy matters and how it changes the legislature's value comparison. Section 5 presents the constructive obstruction result. Section 6 develops the investment implication. Section 7 maps the model onto four constitutional systems. Section 8 discusses extensions and limitations. Section 9 concludes.

2 The Model

2.1 Environment and Timing

I study a two-player game between a legislature A and a court J under certain review.² The legislature writes a statute with two observable characteristics: repairability design $e \geq 0$ and review-relevant burden $x \in [0, \bar{x}]$.

Repairability design captures statute-visible features that make repair more or less feasible if the court opens a remedial window: severability scaffolding, fallback clauses, transition rules, delegated adjustment authority, and built-in revision mechanisms. It is not hidden effort, generic legislative quality, or political competence. It is the observable design of the enacted text.

The timing is:

1. A chooses repairability design e ;
2. A chooses burden x , taking e as sunk;
3. review state σ is realized from a known distribution;
4. J chooses a remedy $r \in \{U, C, S\}$;
5. if C is chosen, A chooses repair intensity ρ ;
6. terminal payoffs are realized.

The parameter $\mu > 0$ captures the jurisdiction-level environment that converts a court-opened repair window into a timely, durable fix conditional on e . It is not a filing parameter, a generic state-capacity index, or a synonym for judicial strength.

Primitives. The legislature's policy-value function $v: [0, \bar{x}] \rightarrow \mathbb{R}$ is strictly increasing with $v(0) = 0$. The parameter $\beta > 0$ is the baseline payoff from an upheld statute. The post-repair benefit function b and repair cost function c govern the continuation stage (Section 2.3). The design cost function $\phi: [0, \infty) \rightarrow [0, \infty)$ with $\phi(0) = 0$ captures the cost of embedding repair-facilitating features in the enacted text (Section 6).

²I condition on review to isolate the remedial-menu mechanism. Endogenizing the challenge decision would add a separate litigation-access margin; Appendix B shows that the main results extend qualitatively under exogenous probabilistic review.

Assumption 1 (Maintained parameters). $0 < h < \bar{x}$ and $\beta + v(h) > 0$.

Assumption 2 (Repair technology). *The repair technology is governed by $\theta(\sigma, e; \mu)$ with $\theta_e > 0$, and repairability design e enters the model only through the contingent-remand channel.*

2.2 Court Objective and Remedy Choice

The court minimizes an explicit loss function:

$$L_J(U \mid x, \sigma) = x, \quad L_J(S \mid x, \sigma) = h, \quad L_J(C \mid x, \sigma, e; \mu) = \kappa + Y_C(x, \sigma, e; \mu), \quad (1)$$

where $h > 0$ is the disruption cost of strike and $\kappa > 0$ is the institutional cost of using the conditional remedy. The normalization $L_J(U \mid x, \sigma) = x$ fixes the scale of the burden dimension without requiring the court and the legislature to value the statute on the same scale, because the legislature's payoff runs through $v(x)$ with $v'(x) > 0$ rather than through x directly.

The court chooses C only under strict improvement:

$$\kappa + Y_C(x, \sigma, e; \mu) < \min\{x, h\}. \quad (2)$$

Otherwise it chooses the better benchmark remedy, with ties going to benchmark remedies and U beating S at $x = h$.

2.3 Continuation Stage

After C , the legislature chooses repair intensity $\rho \in [0, x]$ to maximize a continuation payoff

$$u_A^C(x, \rho, \sigma, e; \mu) = \beta + b(x, \rho, \sigma, e; \mu) - c(\rho, x, \sigma, e; \mu), \quad (3)$$

where $b_x > 0$, $b_\rho \leq 0$, $c_\rho > 0$, and $c_{\rho\rho} \geq 0$. If the optimal repair leaves the legislature worse off than zero, the statute lapses at the end of the remedial window. Let $y(x, \rho, \sigma, e; \mu)$ denote the

terminal burden after repair. Define the reduced-form continuation objects:

$$\begin{aligned}\widetilde{W}_C(x, \sigma, e; \mu) &:= u_A^C(x, \rho^*, \sigma, e; \mu), \\ W_C(x, \sigma, e; \mu) &:= \max\{0, \widetilde{W}_C(x, \sigma, e; \mu)\},\end{aligned}\tag{4}$$

$$Y_C(x, \sigma, e; \mu) := \begin{cases} y(x, \rho^*, \sigma, e; \mu) & \text{if } \widetilde{W}_C > 0, \\ 0 & \text{if } \widetilde{W}_C \leq 0, \end{cases}\tag{5}$$

where $\rho^* = \rho^*(x, \sigma, e; \mu)$ is any maximizer, y is the terminal burden after repair with $y_x > 0$ and $y_\rho < 0$, and the lapse convention keeps failed salvage distinct from successful repair. The lapse convention (truncating W_C at zero) captures situations in which the legislature's repair attempt fails to produce a viable statute within the remedial window, so the statute expires rather than continuing in its original form. This maps onto institutional practice in several jurisdictions: in Canada, a suspended declaration of invalidity means the impugned provision ceases to have legal effect at the end of the suspension period if Parliament has not enacted a replacement; the legislature is left with no statute rather than with the original (now invalid) law. The convention ensures that lapse is weakly worse than strike for the legislature, since both yield a zero payoff, but lapse is distinct from successful repair, which yields positive continuation value.

2.4 Legislative Payoffs

The legislature receives

$$\beta + v(x) \quad \text{under } U, \quad 0 \quad \text{under } S, \quad W_C(x, \sigma, e; \mu) \quad \text{under } C.$$

The key discipline is that e matters only through the continuation technology: it does not directly improve the statute absent review.

For exposition, Sections 3 through 5 hold $e = \bar{e}$ fixed and analyze burden choice. Section 6 reopens endogenous design.

3 Binary Benchmark

Under binary review, the court chooses only between U and S , yielding the value function (I suppress e and μ from V_B because neither affects the binary payoff; Theorem 2(a) makes this formal)

$$V_B(x) = \begin{cases} \beta + v(x) & \text{if } x \leq h, \\ 0 & \text{if } x > h. \end{cases} \quad (6)$$

Proposition 1 (Binary benchmark). *Under Assumption 1, $X_B^* = \{h\}$.*

Proof. On $[0, h]$, V_B is increasing through v , so its maximum is at h . On $(h, \bar{x}]$, $V_B = 0 < \beta + v(h)$. $\square$

The binary benchmark delivers a sharp prediction: the legislature drafts up to the court's strike threshold and no further. The statute extracts the maximum policy value consistent with surviving review. This result is the standard starting point for the shadow-of-review literature (Stephenson, 2004; Rogers and Vanberg, 2002, 2007), and it holds without any structure on the repair technology or the review state because the conditional remedy plays no role. The benchmark also previews the investment contrast developed in Section 6: by Assumption 2, binary review excludes the only channel through which e operates, so the binary benchmark contains no behavioral role for e at all.

4 Where the Middle Remedy Matters

Now add C to the judicial menu and hold $e = \bar{e}$ fixed. The court-relevant regions are

$$R_C^U(\sigma, \bar{e}; \mu) := \{x \leq h : \kappa + Y_C(x, \sigma, \bar{e}; \mu) < x\},$$

$$R_C^S(\sigma, \bar{e}; \mu) := \{x \geq h : \kappa + Y_C(x, \sigma, \bar{e}; \mu) < h\}.$$

On R_C^U , the conditional remedy displaces uphold: the legislature's payoff changes from $\beta + v(x)$ to W_C . On R_C^S , it displaces strike: the payoff changes from 0 to W_C . The first channel can discipline

by reducing the value of statutes that would otherwise survive; the second can insure by creating positive value for statutes that would otherwise be invalidated.

Define the economically relevant salvage subset $R_{C,+}^S(\sigma, \bar{e}; \mu) := \{x \in R_C^S : W_C(x, \sigma, \bar{e}; \mu) > 0\}$ and the ex ante extended-regime value

$$V_E(x; \mu, \bar{e}) = \begin{cases} \beta + v(x) + \mathbb{E}_\sigma[(W_C - \beta - v(x))\mathbf{1}\{x \in R_C^U\}] & \text{if } x \leq h, \\ \mathbb{E}_\sigma[W_C \cdot \mathbf{1}\{x \in R_C^S\}] & \text{if } x > h. \end{cases}$$

The menu wedge $\Delta(x; \mu) := V_E(x; \mu, \bar{e}) - V_B(x)$ records where the remedial menu reprices legislative risk.

Theorem 1 (Middle-remedy characterization). *Fix $e = \bar{e}$ and $\mu > 0$.*

- (a) *Outside $R_C(\sigma, \bar{e}; \mu) := R_C^U \cup R_{C,+}^S$ state by state, $V_E = V_B$. Only policies in R_C^U or R_C^S are affected by the middle remedy.*
- (b) *If $\Pr_\sigma[x \in R_{C,+}^S(\sigma, \bar{e}; \mu)] > 0$ for some $x > h$, then some statute with zero binary-review value acquires strictly positive value under conditional review.*

Part (a) establishes the “where” of the middle remedy: it can only matter where the court actually uses it. Outside the conditional-remedy regions, the legislature’s value is unchanged from the binary benchmark, regardless of the repair technology or the repair-productivity parameter. This is intuitive: a remedy that the court never selects cannot reshape legislative incentives. Stating it formally delineates the boundary between statutes that are affected by the menu expansion and those that are not.

Part (b) is the salvage result: if the court ever opens a repair window on the strike side and the legislature can extract positive continuation value from the repair opportunity, then some statute that would be worthless under binary review acquires positive value under conditional review. This is the insurance channel. The legislature can draft aggressively knowing that even if the statute exceeds the court’s strike threshold, the conditional remedy may preserve a positive payoff.

Together, these two parts explain why the middle remedy cannot be summarized by a single directional slogan. The same menu expansion disciplines some statutes by displacing survival with

conditional repair (which may yield lower payoff than outright uphold) and insures others by displacing strike with conditional repair (which yields positive payoff where the binary benchmark yields zero). Whether the net effect pushes the legislature toward more or less aggressive drafting depends on the relative strength of these two channels, which in turn depends on the specific structure of the repair technology and the distribution of review states. The next section shows that this dependence is not merely a matter of degree but produces qualitative non-monotonicity.

5 The Directional Question

Does higher repair productivity make the legislature more or less aggressive? Under binary review, the question does not arise: the benchmark optimum is at the strike threshold regardless of the repair environment. Under conditional review, a reader might expect a single threshold separating regimes where the conditional remedy softens discipline from regimes where it sharpens discipline. That expectation is false in general.

Proposition 2 (Obstruction). *Fix $e = \bar{e}$. There exists an admissible environment in the maintained nonlinear witness class such that*

$$X^*(\mu) := \arg \max_{x \in [0, \bar{x}]} V_E(x; \mu, \bar{e})$$

fails any one-cutoff monotonic pattern in μ . Specifically, for the environment

$$h = 1, \quad \bar{x} = \frac{3}{2}, \quad \kappa = \frac{2}{5}, \quad \beta = 1, \quad v(x) = \frac{x}{2}, \quad (7)$$

with $\sigma \in \{g, b\}$, $\Pr(g) = 3/10$, and witness primitives

$$\tau(g, \bar{e}) = 10, \quad a(g, \bar{e}) = 5, \quad \tau(b, \bar{e}) = 3, \quad a(b, \bar{e}) = \frac{1}{100}, \quad (8)$$

the optimizer correspondence satisfies

$$X^*\left(\frac{1}{4}\right) = \left\{\frac{3}{2}\right\}, \quad X^*\left(\frac{11}{20}\right) = \left\{\frac{825}{914}\right\}, \quad X^*\left(\frac{7}{10}\right) = \left\{\frac{525}{457}\right\}.$$

The pattern is aggressive, then nonaggressive, then aggressive again as repair productivity rises. No threshold $\hat{\mu}$ can make aggressive optimality equivalent to either $\mu \geq \hat{\mu}$ or $\mu \leq \hat{\mu}$. Because all comparisons in the counterexample are strict, the same pattern persists on small neighborhoods of the three reference values.

Proof sketch. The counterexample uses the maintained nonlinear witness formulas

$$Y_C(x, \sigma, \bar{e}; \mu) = \max\{0, x - \mu\tau(\sigma, \bar{e})\sqrt{x}\}$$

and

$$W_C(x, \sigma, \bar{e}; \mu) = \max\left\{0, 1 + \frac{x}{2} - \frac{a(\sigma, \bar{e})}{\mu\tau(\sigma, \bar{e})}x^2\right\}.$$

For $\mu \in \{1/4, 11/20, 7/10\}$, the court uses U on $[0, 2/5]$ and C on $(2/5, 3/2]$ in both states, so the ranking change is driven entirely by the shape of the repaired continuation value, not by a different top-level remedy map. The full branch calculations are in Appendix C. $\square$

Three mechanisms drive the non-monotonicity, and each is characteristic of the conditional-remedy environment rather than an artifact of the specific parameter values.

First, truncation kinks arise at low repair productivity ($\mu = 1/4$). The good-state repaired payoff is truncated to zero on the far right of the burden range, while the bad-state payoff remains increasing. That truncation leaves a rising right tail composed predominantly of the bad-state continuation value, making the maximum feasible burden uniquely optimal.

Second, left-envelope re-expansion operates at intermediate repair productivity ($\mu = 11/20$). As repair productivity rises, the two-state repaired quadratic shifts upward enough to create an interior maximizer below the strike threshold. Higher repair productivity improves the value of nonaggressive repaired statutes as well as aggressive ones, and at this parameter value the nonaggressive option

dominates.

Third, attainment failure reinforces the intermediate case. The best aggressive value at $\mu = 11/20$ is a left-limit at the strike threshold, not an attained interior maximum. The supremum over aggressive statutes falls short of the nonaggressive maximizer.

At higher repair productivity ($\mu = 7/10$), the repaired-region vertex crosses back above the strike threshold, restoring aggressive optimality. The resulting non-monotonicity is not a failure of proof technique; it is a consequence of the continuation-value structure that the conditional remedy introduces into the legislature's problem. Appendix D records the local boundary diagnostics around $x = h$ that formalize the one-sided level comparisons underlying these mechanisms.

Remark 1 (Attainment and the role of the strict-improvement rule). At $\mu = 11/20$, the best aggressive value is a boundary limit (a supremum at $x \downarrow h$) rather than an attained interior maximum. A natural question is whether this attainment failure is essential to the non-monotonicity or merely reinforces it. Two observations clarify the role of attainment. First, in practice a legislature can draft at $x = h + \varepsilon$ for arbitrarily small $\varepsilon > 0$, so the distinction between an attained maximum and a boundary limit is economically negligible. Second, and more importantly, the truncation-kink and left-envelope mechanisms are independent of the attainment question: they arise from the shape of the continuation value on the nonaggressive side and from the interaction between state-specific truncation points on the aggressive side. Even if the aggressive-side supremum were attained (for instance, under a weak-improvement rule that allows C when the court is indifferent, or under a closed remedy region), the nonaggressive maximizer at $\mu = 11/20$ would still dominate because its value exceeds the aggressive supremum by a strict margin that does not depend on the open/closed convention. The attainment failure is therefore a reinforcing feature of the counterexample rather than a necessary condition for non-monotonicity.

Remark 2 (Genericity). The non-monotonicity is proved for one environment, but the strict inequalities in Proposition 2 imply that it is not isolated. Since the value functions $V_E(x; \mu, \bar{e})$ are continuous in the primitive vector $\pi := (\text{Pr}(g), \tau(g), \tau(b), a(g), a(b), \kappa, \beta, \bar{x}, h)$, and the three value compar-

isions at $\mu \in \{1/4, 11/20, 7/10\}$ are all strict, the set

$$\Pi_{\text{NM}} := \{\pi : X^*(\mu) \text{ exhibits the aggressive-nonaggressive-aggressive pattern}\}$$

is open in the primitive space. An open set in $\mathbb{R}^9$ that contains a point has positive Lebesgue measure. The non-monotone set Π_{NM} therefore has positive Lebesgue measure in the full parameter space, though I do not claim it is generic in the sense of containing an open dense set.

Figure 1 plots the three value functions. Figure 2 maps the aggressive and nonaggressive classification across the (μ, κ) parameter space, showing that the non-monotone pocket is not knife-edge.³

6 Repairability Investment

The obstruction result shows that conditional review changes the level of legislative aggressiveness in a structurally ambiguous way. But the same continuation channel also changes what the legislature builds into the statute before any review occurs. Under binary review, repairability design is irrelevant because there is no repair opportunity. Under conditional review, a well-designed statute may survive where a poorly designed one would lapse. That asymmetry creates a new ex ante design margin.

The logic is straightforward. If the court never opens a repair window, then investing in features that facilitate repair is pure waste. Binary review is exactly such a regime: the court either upholds or strikes, and in neither case does the statute enter a repair process. Any positive design cost therefore pushes the binary optimum to zero investment. But once the court can choose the conditional remedy, the legislature's continuation value depends on how effectively the statute can be repaired, which in turn depends on the design features embedded in the enacted text.

Define the outer objectives

$$\Psi_B(e; \mu) := -\phi(e) + \sup_{x \in [0, \bar{x}]} V_B(x, e; \mu), \quad \Psi_E(e; \mu) := -\phi(e) + \sup_{x \in [0, \bar{x}]} V_E(x, e; \mu).$$

³Appendix C reports a one-at-a-time sensitivity analysis confirming that the pattern persists on small neighborhoods of the baseline witness environment.

Theorem 2 (Repairability investment).

- (a) **Binary dormancy.** $V_B(x, e; \mu) = V_B(x, 0; \mu)$ for all x, e, μ . Hence $e = 0$ is optimal under binary review, and uniquely so if ϕ is strictly increasing.
- (b) **Conditional activation.** If $\Psi_E(\cdot; \mu)$ admits a right derivative at $e = 0$ and $\Psi'_{E,+}(0; \mu) > 0$, then $e = 0$ is not optimal under conditional review. If the outer argmax is nonempty, every optimum satisfies $e_E^*(\mu) > 0$.

Proof sketch. Part (a): e enters only through the contingent-remand channel, which binary review excludes, so V_B is independent of e and the outer problem subtracts $\phi(e)$ from a constant. Part (b): a strictly positive right-hand derivative at zero means moving slightly above zero raises value. No separability, global concavity, or monotone comparative statics in μ are needed. Appendix E gives the full argument. $\square$

The result is a qualitative discontinuity between the two review regimes, not a quantitative one. Under binary review, the marginal benefit of repairability is identically zero regardless of the repair-productivity environment, because the contingent-remand channel through which repairability operates is entirely absent. Under conditional review, the marginal benefit can be strictly positive because the repair channel converts design into continuation value. Within the model, the contrast is sharp: any positive repairability investment that arises under conditional review is attributable to the availability of the conditional remedy itself, not to features of the baseline review environment.

When the benefit and cost pieces are separately right-differentiable at zero, the condition in part (b) reduces to requiring that the marginal benefit of repairability at zero exceeds the marginal design cost at zero:

$$\left. \frac{d}{de} \sup_{x \in [0, \bar{x}]} V_E(x, e; \mu) \right|_{e=0,+} > \phi'_+(0).$$

The left side captures how much a small improvement in repair-facilitating design raises the legislature's best achievable value; the right side captures the marginal cost of that improvement.⁴

I do not claim a monotone comparative static on $e_E^*(\mu)$. The model earns the qualitative result

⁴Appendix F records a constructive multiplicative witness example ($g_M(e) = \sqrt{1+e}$, $\phi(e) = 10^{-3}e$) in which $\Psi'_{E,+}(0; \mu) = 0.00627 > 0$, confirming that the condition is satisfiable.

(conditional review activates the investment margin) but not a general prediction about how optimal investment varies with repair productivity. That silence reflects a genuine limitation: the multiplicative witness class does not by itself imply increasing differences in the reduced-form outer objective, so a Topkis-style monotone selection result would require additional structure beyond what the current model supplies.

7 Institutional Mapping

This section maps the model’s objects onto four constitutional systems that employ conditional remedies, illustrating how the parameter μ varies across jurisdictions and connecting the formal results to observable institutional variation. The model is intentionally stylized (it features a unitary legislature, a scalar burden, a mechanical court, and certain review), so the mappings below should be read as illustrative connections between formal objects and institutional practice rather than as unconditional empirical predictions. The value of the exercise lies in identifying which model parameters are most relevant for each jurisdiction and where the model’s maintained assumptions are most and least plausible.

7.1 Canada: Suspended Declarations

The Canadian Supreme Court issues suspended declarations of invalidity that keep a statute in force for a fixed period, typically 6 to 18 months, while Parliament drafts a replacement. In *Carter v. Canada*, [2015] 1 SCR 331, the Court suspended its declaration on physician-assisted dying for 12 months, later extended to 16; in *Canada (Attorney General) v. Bedford*, [2013] 3 SCR 1101, the Court gave Parliament 12 months to rewrite the criminal provisions on sex work; and in *Corbiere v. Canada*, [1999] 2 SCR 203, the Court granted Parliament 18 months to consult with more than 135 affected groups before amending the Indian Act’s residence requirements. These suspensions map directly onto the model’s conditional remedy C : the court preserves the statute conditional on legislative repair within a specified window. The *Corbiere* suspension, in particular, illustrates how the conditional window can serve a repair-capacity function, giving the legislature time to assemble

the information needed for a durable fix.

The empirical literature on Canada’s “Charter dialogue” supports the model’s core prediction that intermediate remedies channel legislative behavior rather than block it. Hogg and Bushell (1997) found that legislatures responded with new legislation to approximately two-thirds of Charter decisions invalidating laws, and Hogg, Bushell Thornton, and Wright (2007) updated and defended these findings against subsequent critiques. This two-thirds response rate implies that the shadow of conditional review generates a substantial legislative compliance channel, consistent with the model’s prediction that the conditional remedy creates continuation value that the legislature can capture through design effort. Roach (2004) documents the growth of suspended declarations from 14 percent of Charter nullification cases in 1989–93 to 57 percent in 1998–2003, suggesting that courts have progressively expanded the C region over time. In the model’s terms, this expansion intensifies the shadow of conditional review. Bird (2019) offers a cautionary perspective, arguing that overuse of suspended declarations has turned them into a “judicial notwithstanding clause” that undermines rights protection during the suspension period, a moral-hazard concern that maps onto the possibility that expanding C reduces the legislature’s *ex ante* effort to comply with constitutional constraints.

Canada also illustrates the role of κ (institutional cost of the conditional remedy). The doctrinal framework established in *Schachter v. Canada*, [1992] 2 SCR 679, provides clear guidelines for when suspension is appropriate, making the remedy procedurally routine and well understood by legislative counsel. This established framework places Canada in a low- κ environment: the institutional friction of deploying C is small relative to jurisdictions where the conditional remedy’s legal basis is contested.

The investment margin (Theorem 2) implies that, if the right-derivative condition holds in this setting, Canadian legislatures facing suspended declarations would have a positive incentive to draft statutes with built-in fallback provisions, severability scaffolding, and transition rules, precisely because the suspended declaration converts those design features into continuation value. A legislature facing only binary review would have no such incentive.

7.2 Germany: Incompatibility Declarations

Germany’s Federal Constitutional Court (*Bundesverfassungsgericht*) provides a second mapping. Its incompatibility declaration (*Unvereinbarerklärung*) declares a law incompatible with the Basic Law without immediately voiding it, directing the Bundestag to enact a compliant replacement within a specified deadline. This stands in contrast to outright nullification (*Nichtigerklärung*), the binary-review equivalent. The incompatibility route is concentrated in tax law and social welfare law, where immediate nullification would create greater inequality or fiscal chaos (Kommers and Miller, 2012). In the Hartz IV ruling (BVerfGE 125, 175, 2010), for example, the Court declared the method for calculating subsistence benefits incompatible with human dignity and directed the Bundestag to enact a constitutionally adequate replacement; the legislature complied within the specified window. The domain concentration maps onto the model’s parameter h (disruption cost of strike): the Court uses C precisely when h is large, consistent with the model’s prediction that the conditional remedy is used when the cost of outright strike is prohibitive.

Germany plausibly represents a high- μ environment. Vanberg (2005) provides the key compliance mechanism: legislative compliance with Constitutional Court deadlines is strongly conditioned by public opinion transparency. When the constitutional issue is salient and public opinion is firm, the Bundestag is significantly more likely to comply. Kommers and Miller (2012) document the same pattern across decades of jurisprudence. In the model, the credibility of C depends on an enforcement mechanism that makes legislative defection from court-imposed deadlines costly; Vanberg’s public-opinion channel provides exactly this, grounding the model’s maintained assumption that conditional remedies are incentive-compatible in high-capacity jurisdictions.

The German case also speaks to the model’s treatment of the court as a mechanical actor. Krehbiel (2021) shows that the *Bundesverfassungsgericht* uses incompatibility declarations strategically: they are issued more frequently when the incumbent government is electorally weak, because delay raises the probability that a more amenable successor government will implement the court’s preferred policy. In the model, the court’s remedy regions are exogenous; Krehbiel’s finding that real courts choose C versus S endogenously connects to the strategic-court extension discussed in Section 8. The mechanisms identified in Proposition 2 survive this extension because they arise from

the geometry of the legislature’s continuation value rather than from the court’s decision rule.

The obstruction result implies that, conditional on the maintained assumptions, even in a high- μ jurisdiction the directional effect of the incompatibility remedy on legislative aggressiveness cannot be signed in advance. The investment margin (Theorem 2) similarly implies that German legislators would have reason to incorporate repair-facilitating features into statutory text that they would not include under pure nullification.

7.3 Indonesia: Conditionally Constitutional Rulings

Since its establishment in 2003, Indonesia’s Constitutional Court (*Mahkamah Konstitusi*, MK) has issued dozens of conditionally constitutional rulings (*putusan konstitusional bersyarat*) that preserve a statute subject to a specified interpretive or legislative fix.⁵ The MK’s remedial practice has evolved through distinct phases of judicial leadership. Hendrianto (2018) identifies three models (prudential-minimalist under the early Court, aggressive-bold during the activist period, and a more deferential posture thereafter) and Butt (2015) corroborates the shift toward expansive remedial practice. In the model, the court is a mechanical actor; the Indonesian experience illustrates that the effective scope of conditional review is partly endogenous to judicial strategy, a complication the model’s maintained assumption abstracts from (as acknowledged in Section 8).

Quantitative compliance data place Indonesia at the low end of the μ spectrum. Approximately 50 percent of MK decisions from 2013 to 2018 were fully complied with; roughly one-third were outright ignored (Hendrianto, 2018). Butt and Murharjanti (2022) provide an important nuance: many decisions coded as “non-compliance” in raw tallies actually require no legislative response because they are self-executing or require only regulatory adjustments. This complicates simple measurement of μ ; the model treats repair productivity as a scalar, but the Indonesian evidence suggests it is domain-specific and partly endogenous to remedy design.

The Job Creation Omnibus Law case illustrates the compliance problem at its sharpest. In De-

⁵On the MK’s procedural framework for conditional rulings, see Constitutional Court of Indonesia, Regulation No. 06/PMK/2005 on Procedural Guidelines; Wicaksono and Rahman (2022) discuss the MK’s evolving remedial practice.

cision No. 91/PUU-XVIII/2020 (November 2021), the MK declared the massive omnibus law conditionally unconstitutional on procedural grounds and gave the legislature two years to remedy the defects. The government responded by re-enacting the law via emergency regulation (*Peraturan Pemerintah Pengganti Undang-Undang*, Perppu), circumventing the Court’s conditional window entirely. In the model’s terms, this amounts to setting $\mu = 0$ through an institutional workaround: rather than using the repair channel the Court opened, the executive substituted an alternative legislative pathway that bypassed the conditional window. The case also illustrates high h : the omnibus law was so large and politically consequential that immediate nullification was effectively off the table, which is precisely the condition under which the model predicts that the conditional remedy will be deployed.

The obstruction result (Proposition 2) cautions against coding Indonesian conditional rulings as uniformly lenient or uniformly disciplinary. Whether these rulings soften or sharpen legislative incentives depends on μ , which varies across statutory domains and legislative periods. Empirical work on Indonesia’s conditional remedy would benefit from identifying domain-specific proxies for repair productivity rather than treating the remedy as a single treatment.

7.4 South Africa: Reading-In with Deferral

South Africa’s Constitutional Court operates under Section 172 of the Constitution, which grants expansive remedial discretion including the power to make “any order that is just and equitable.” Klug (2000) documents how these broad remedial powers emerged from the constitutional negotiation process: the drafters deliberately equipped the Court with a flexible toolkit to manage the transition from apartheid, giving the judiciary access to the full $\{U, C, S\}$ menu with minimal institutional friction. Roux (2013) shows that under Chief Justice Chaskalson, the Court balanced principled jurisprudence with political survival under ANC governance, deploying suspended declarations and reading-in strategically to build institutional legitimacy while advancing constitutional rights. In the model’s framework, the combination of broad constitutional authorization and an established doctrinal tradition places South Africa in a low- κ region: the institutional cost of deploying conditional remedies is small because the legal infrastructure for doing so is constitutionally embedded.

The leading example of South Africa’s conditional remedy is *Minister of Home Affairs v. Fourie* [2005] ZACC 19. The Court declared the heterosexual-only definition of marriage unconstitutional, suspended the declaration for 12 months, and specified that if Parliament failed to act, the words “or spouse” would automatically be read into the Marriage Act. Parliament responded within the window by enacting the Civil Union Act. The automatic reading-in provision represents a strong-form conditional remedy in which the court specifies the repair outcome that will obtain absent legislative action. In the model’s terms, this amounts to a conditional remedy with a known outside option, suggesting that an extension incorporating a court-specified default could further sharpen the legislature’s ex ante incentive to design the statute for repair rather than risk a court-imposed fallback.

The Court’s doctrine also reveals an information-structure constraint absent from the model. In *Dawood v. Minister of Home Affairs* [2000] ZACC 8, the Court held that reading-in is appropriate only when the court can define “with sufficient precision” how the law must be amended; where multiple legitimate legislative options exist, the court defers via suspended declaration.⁶ The model assumes that the conditional remedy has a well-defined repair outcome; the *Dawood* doctrine identifies when that assumption binds. When courts cannot specify the repair (because the constitutional defect admits multiple corrective pathways), they default to the binary $\{U, S\}$ menu. This is a testable implication: conditional remedies should cluster in domains where the constitutionally required fix is determinate.

Variation in compliance speed across cases within South Africa supports the cross-cutting claim that μ varies by statutory domain, not only across jurisdictions. In *Teddy Bear Clinic v. Minister of Justice* [2013] ZACC 35, the Court suspended its declaration for 18 months rather than reading in, because the impugned provisions were so interlinked with other statutory sections that no single textual correction could remedy the defect, a high- h configuration where the disruption cost of immediate nullification was prohibitive. By contrast, after *Masiya v. Director of Public Prosecutions* [2007] ZACC 9, Parliament enacted the Criminal Law (Sexual Offences and Related Matters)

⁶See also *National Coalition for Gay and Lesbian Equality v. Minister of Justice* [1998] ZACC 15, where the Court applied this precision threshold.

Amendment Act within months, reflecting a domain where both repair productivity and political will were high. This within-jurisdiction variation illustrates the model's prediction that the same formal remedy type can produce different incentive effects depending on the underlying institutional parameters.

7.5 Repair Productivity Across Jurisdictions

The cross-country variation suggests that μ is not a theoretical abstraction but an institutionally grounded parameter that admits empirical proxies. Three observable features help locate a jurisdiction in the μ space:

1. **Compliance rates with court-imposed deadlines.** The share of conditional remedies in which the legislature completes a repair within the specified window directly measures the productivity of the repair channel. Canada's legislature responded with new legislation to approximately two-thirds of Charter invalidations (Hogg and Bushell, 1997; Hogg, Bushell Thornton, and Wright, 2007). Germany's compliance rate is high, particularly when the constitutional issue is publicly salient (Vanberg, 2005). Indonesia's is substantially lower: roughly 50 percent of MK decisions were fully complied with between 2013 and 2018, with approximately one-third outright ignored (Hendrianto, 2018; Butt and Murharjanti, 2022). South Africa's compliance varies by remedy type, with rapid legislative response in some domains (*Masiya*) and protracted delay in others. This variation maps onto the model's prediction that the same formal remedy type can produce different incentive effects at different values of μ .
2. **Legislative drafting capacity.** The number of dedicated legislative drafting staff, the availability of specialized offices of legislative counsel, and the extent of legislative support infrastructure all affect how efficiently a legislature converts a court-opened window into a durable fix. The Bundestag's extensive committee staffing and specialized drafting offices contrast with the more limited resources available to the Indonesian DPR, where Hendrianto (2018) documents that drafting bottlenecks have contributed to non-compliance with MK deadlines. Canada's system of legislative counsel offices, which routinely advise on Charter compliance during the drafting process, represents an intermediate case that supports high μ through ex

ante institutional design.

3. **Average time to legislative response.** The elapsed time between a conditional ruling and the enactment of responsive legislation provides a continuous proxy for μ . Canadian suspended declarations typically carry 12–18 month windows, though Bird (2019) notes that overuse has stretched some timelines considerably. German response times vary by domain, with tax and social welfare cases often resolved faster than structural reforms. Indonesia’s two-year windows are frequently unmet. South Africa’s 12–18 month suspensions produce mixed compliance. Shorter response times, controlling for the complexity of the required repair, indicate higher repair productivity.

The four jurisdictions also vary along two additional model parameters. The institutional cost of deploying conditional remedies (κ) is low in Canada and South Africa, where established doctrinal frameworks (*Schachter* and Section 172, respectively) make the conditional remedy a routine part of the judicial toolkit. Germany’s κ is moderate: the incompatibility declaration is one of several available remedial routes, and its deployment involves a more structured doctrinal calculus (Kommers and Miller, 2012). Indonesia’s κ is uncertain, because the MK’s authority to issue conditional rulings has itself been contested (Butt, 2015). The disruption cost of strike (h) also varies systematically: Germany’s concentration of incompatibility declarations in tax and social welfare law reflects domains where immediate nullification would create fiscal chaos or inequality, precisely the high- h configurations where the model predicts the conditional remedy will be deployed. Indonesia’s Job Creation Law case represents an extreme h environment where the sheer scale of the omnibus legislation made nullification effectively impossible.

The Indonesia-Germany contrast is particularly informative because both jurisdictions employ conditional remedies but sit at opposite ends of the μ spectrum. Empirical identification of the non-monotonicity prediction (Proposition 2) requires variation in μ that is plausibly exogenous to the legislature’s burden choice; cross-country differences in legislative capacity and compliance norms are natural candidates. The additional variation in κ and h across jurisdictions provides further leverage: the model predicts that conditional remedies will be deployed more frequently in high- h domains (where strike is costly) and low- κ jurisdictions (where the institutional infrastructure for

conditional review is well established), and these predictions are testable against the cross-national patterns documented above.

8 Discussion

Strategic courts. The court in the model minimizes a loss function mechanically. In the formal judicial politics literature, courts are often modeled as strategic actors with preferences over policy outcomes and institutional capital (Stephenson, 2004; Bueno de Mesquita and Stephenson, 2002). A forward-looking court that anticipates the legislature’s burden response could alter the remedy regions endogenously: if the court foresees that expanding the conditional-remedy region will induce more aggressive drafting, it might restrict its use of C to counteract that effect. Whether this strategic adjustment eliminates the non-monotonicity depends on whether the court can commit to a remedy rule *ex ante*. Under commitment, the court’s optimal rule would internalize the legislature’s response, potentially smoothing the non-monotone pocket. Without commitment, the court optimizes state by state, and the three mechanisms identified in Section 5 (truncation kinks, left-envelope re-expansion, and attainment failure) arise from the geometry of the continuation value rather than from the court’s decision rule, so they plausibly persist as long as the conditional remedy remains in the court’s menu, though a formal proof under endogenous remedy regions is an open problem. A full treatment of the strategic-court extension is beyond the scope of this paper, but the key point is that the mechanisms are characteristic of the conditional-remedy environment rather than artifacts of the mechanical court assumption.⁷

Repeated interactions and learning. The model is one-shot: the legislature drafts a single statute under the shadow of a single review. In practice, legislatures face judicial review repeatedly and can

⁷The “mechanical court” language is shorthand for sequential best response at the realized remedy node under the maintained no-commitment timing. At each (x, σ) , the court’s loss-minimization problem is single-valued given the tie-breaking rule in Section 2.2, so the court’s realized remedy choice is a continuation best response. The analysis throughout the paper can therefore be read as characterizing on-path play under sequential rationality rather than as imposing a behavioral restriction distinct from equilibrium reasoning.

learn from past conditional remedies about the court’s behavior and their own repair productivity. Repeated interaction could introduce reputation effects, precedent dynamics, and Bayesian updating on μ . Whether the non-monotonicity survives depends on the legislature’s learning speed relative to the persistence of the institutional parameters. If μ is stable across legislative periods (reflecting structural features like drafting capacity and compliance norms), the one-shot analysis applies period by period. If μ is learned over time, the legislature’s ex ante problem in early periods incorporates option value from future learning, which could smooth or amplify the non-monotone pattern. I leave the dynamic extension as an open question; the one-shot model isolates the remedial-menu mechanism without conflating it with learning or reputation channels.

Additional modeling limitations. Several maintained assumptions deserve explicit acknowledgment. First, the legislature is modeled as a unitary actor. Real legislatures involve multiple veto players, committee structures, and coalition dynamics that could interact with the conditional-remedy channel in ways the model does not capture. Second, statutory burden is a scalar; real statutes are multidimensional objects, and courts may evaluate different dimensions under different remedial standards. Third, the court observes the review state perfectly. Judicial error (either in assessing the state or in calibrating the remedy) could alter the legislature’s ex ante calculus, particularly if the probability of erroneous conditional remedies varies with burden. Fourth, the binary state space ($\sigma \in \{g, b\}$) is sufficient for the counterexample but limits the richness of state-dependent truncation patterns; a richer state space could generate additional non-monotone pockets or smooth existing ones. Fifth, the behavior of the non-monotone pocket as $\kappa \rightarrow 0$ (when the institutional cost of deploying the conditional remedy vanishes) is not characterized; whether the pocket persists, expands, or collapses in that limit is an open question. Sixth, the optimizer correspondence $X^*(\mu)$ is generically single-valued in the witness class, but multiplicity at boundary parameter values is resolved by convention rather than by economic argument. Finally, while the non-monotone pocket in Proposition 2 is robust to local perturbation (Remark 2 and Table 1), its economic magnitude (the difference between the aggressive and nonaggressive optima) is modest in the witness environment; whether comparable effects arise at empirically relevant scales is an open question.

Certain review as a benchmark. The main text conditions on certain review to isolate the remedial-menu mechanism. This assumption is most natural in jurisdictions with abstract review, where a constitutional challenge can be initiated automatically or by a designated institutional actor regardless of a specific case or controversy. Germany’s abstract *Normenkontrolle* (Article 93(1)(2) of the Basic Law), France’s *question prioritaire de constitutionnalité* (QPC), and several Latin American systems with automatic review of organic laws all approximate the certain-review benchmark. Even in concrete-review systems, the probability of challenge for salient legislation approaches one for politically contested statutes. Appendix B shows that non-monotonicity extends qualitatively to probabilistic review, though the exact switch points shift.

9 Concluding Remarks

Expanding the judicial menu from binary review to a regime with a conditional remedy changes the shadow of review in two linked ways. It obstructs any general global sign prediction for the legislature’s burden choice, and it creates a new *ex ante* reason to design statutes for repair. The first effect is captured by a constructive counterexample inside the maintained class; the second by the investment result. Together they imply that conditional remedies may be more fruitfully analyzed as institutions that reshape both legislative discipline and statutory design rather than as simple soft or hard variants of binary review.

The obstruction result has a direct implication for empirical work. The direction of discipline cannot be answered by coding conditional remedies as “more lenient” or “more disciplinarian” in the abstract. The answer depends on the geometry of the repaired continuation value, the location of remedy regions, the distribution of review states, and the productivity of repair. Empirical work that wants to sign the discipline effect would benefit from identifying the primitives that govern these objects rather than treating the remedial menu as a single monotone treatment variable. The institutional proxies for μ identified in Section 7 (compliance rates, drafting capacity, and response times) provide a starting point for such identification.

I have not determined the sign of the welfare effect. The court’s burden-minimization objective

and the legislature’s policy-value objective are kept separate throughout the analysis, so a welfare statement would require a social welfare function or Pareto criterion that the current model does not supply. Appendix G shows that in the witness economy, conditional review benefits the court and harms the legislature at all three reference values, so neither regime Pareto-dominates. The non-monotonicity of the obstruction result makes this gap especially consequential: whether a directional change in legislative aggressiveness is welfare-improving depends on normative weights that are outside the model.

Three limitations remain explicit. First, the model assumes certain review rather than strategic litigant entry (Appendix B relaxes this partially; Section 8 discusses why certain review is the natural benchmark for abstract-review jurisdictions). Second, I do not sign any general relation between aggressiveness and statutory quality. Third, the analysis is silent on a global comparative static for $e_E^*(\mu)$; the model earns the marginal investment implication, not a monotone investment result.

Several avenues remain open. On the theory side, deriving general sufficient conditions under which non-monotonicity arises would move the result from a possibility theorem to a characterization. Natural extensions include endogenizing the review probability, formalizing the strategic-court extension, and incorporating repeated interactions. On the empirical side, testing the non-monotonicity prediction requires identifying primitives that govern repair productivity. Jurisdiction-level variation in repair-window rules, such as the length and conditions of Canadian suspended declarations versus Indonesian conditionally constitutional rulings, may provide the relevant exogenous variation.

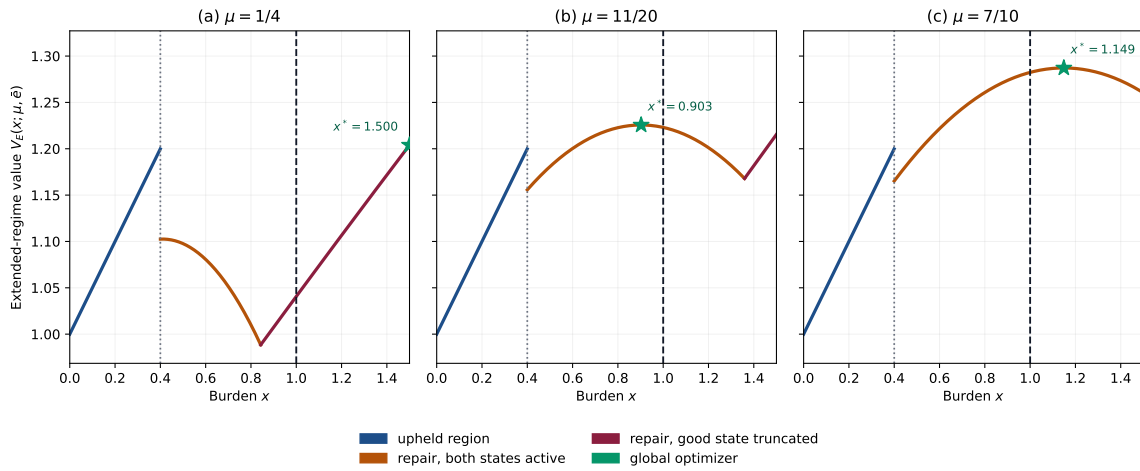


Figure 1: Extended-regime value at three levels of repair productivity.

Notes: The figure plots $V_E(x; \mu, \bar{e})$ at $\mu = 1/4$ (solid), $\mu = 11/20$ (dashed), and $\mu = 7/10$ (dotted), illustrating the aggressive-nonaggressive-aggressive pattern of Proposition 2. The global maximizer lies to the right of $h = 1$ at $\mu = 1/4$ and $\mu = 7/10$ but to the left at $\mu = 11/20$. Vertical dashed line: strike threshold $h = 1$. Horizontal axis: burden $x \in [0, 3/2]$; vertical axis: $V_E(x; \mu, \bar{e})$. Parameter values from Proposition 2: $h = 1, \bar{x} = 3/2, \kappa = 2/5, \beta = 1, v(x) = x/2, \Pr(g) = 3/10$. Source: authors' calculations from the witness environment.

Alt text: Three curves plotting the legislature's value function against statutory burden. The solid curve ($\mu = 1/4$) peaks to the right of the strike threshold; the dashed curve ($\mu = 11/20$) peaks to the left; the dotted curve ($\mu = 7/10$) peaks to the right again. The shifting peak locations illustrate the aggressive-nonaggressive-aggressive pattern.

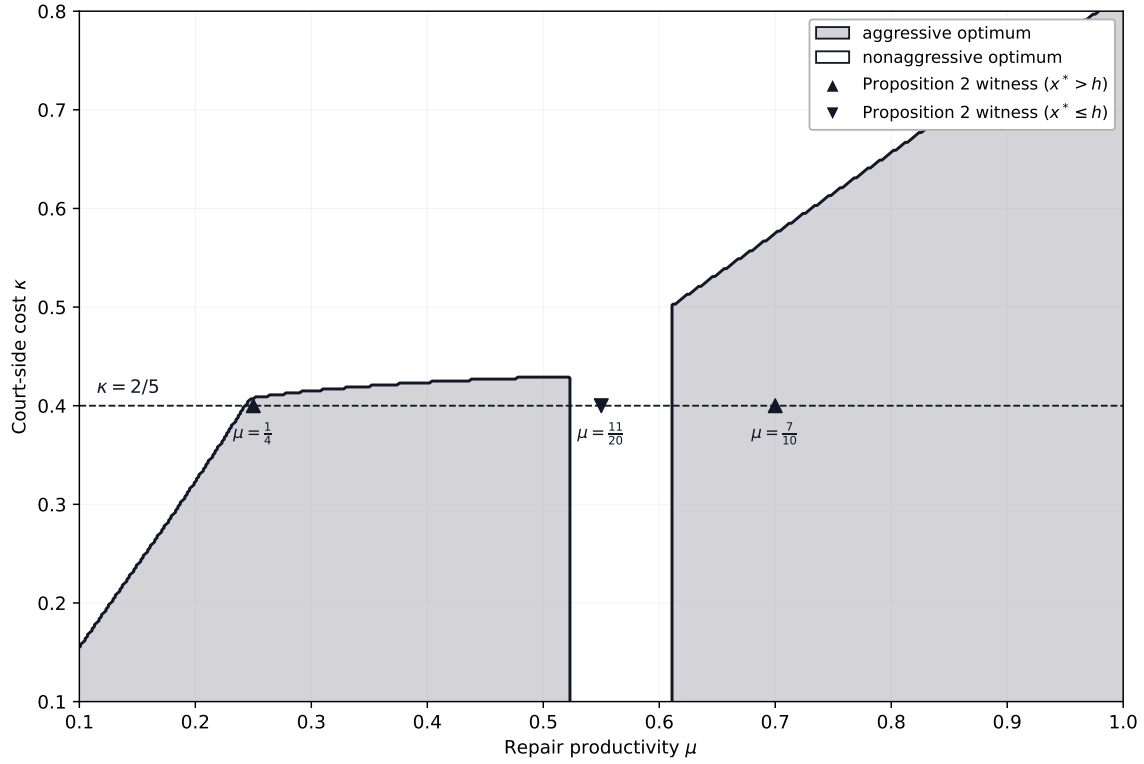


Figure 2: Aggressive vs. nonaggressive optimum in (μ, κ) space.

Notes: The figure maps the global optimum type across the (μ, κ) parameter space for the witness environment of Proposition 2. The non-monotone pocket along the $\kappa = 2/5$ cross-section confirms that non-monotonicity is not a knife-edge artifact. Shaded region: aggressive ($X^* \cap (h, \bar{x}] \neq \emptyset$). Unshaded region: nonaggressive ($X^* \subset [0, h]$). Dashed line: $\kappa = 2/5$. Triangles mark the three Proposition 2 witness values at $(\mu, \kappa) = (1/4, 2/5)$, $(11/20, 2/5)$, and $(7/10, 2/5)$. Horizontal axis: repair productivity μ ; vertical axis: institutional cost κ . All primitives held at Proposition 2 baseline ($h = 1$, $\bar{x} = 3/2$, $\beta = 1$, $v(x) = x/2$, $\Pr(g) = 3/10$, witness curvatures as in (8)). Source: authors' calculations.

Alt text: A two-dimensional heat map with repair productivity on the horizontal axis and institutional cost on the vertical axis. Shaded areas indicate parameter combinations where the legislature drafts aggressively; unshaded areas indicate nonaggressive drafting. Along the horizontal dashed line at $\kappa = 2/5$, the shading alternates from aggressive to nonaggressive and back to aggressive as μ increases, confirming the non-monotone pattern.

Appendix A Witness-Class Definitions and Verification

This appendix records the reduced-form witness families used in Appendix C and Appendix F. They are admissible instantiations of the continuation objects (4)–(5); they are not additional global assumptions on the model.

Nonlinear fixed- $\bar{e}$ witness class. For the burden-choice problem, hold $e = \bar{e}$ fixed and choose positive state-specific primitives $\tau(\sigma, \bar{e}) > 0$ and $a(\sigma, \bar{e}) > 0$. Define

Remark 3 (Primitive foundations of the witness class). The witness formulas (9)–(10) arise from the continuation primitives (b, c, y) in (3) under two conditions: (i) the benefit function b is affine in x (specifically, $b(x, \rho, \sigma) = \beta + v(x) - f(\sigma)\rho^2/x$ for some state-specific $f(\sigma) > 0$), so that the legislature’s pre-repair benefit is linear in the burden; and (ii) the repair cost c is quadratic and convex in ρ with a coefficient that depends on the repair technology parameter θ .

Under these two conditions, optimal repair intensity ρ^* is interior, producing a quadratic residual in x after substitution. The parameter $\tau(\sigma, \bar{e})$ governs the effectiveness of repair in reducing judicial loss (higher τ means repair is more productive at reducing the post-repair burden), while $a(\sigma, \bar{e})$ governs the curvature of the legislature’s net continuation payoff (higher a means the continuation value declines more steeply with burden). These are not additional assumptions on the model; they are sufficient conditions on the primitives that generate the reduced-form witness structure.

$$\theta(\sigma, \bar{e}; \mu) := \mu\tau(\sigma, \bar{e}),$$

and let the continuation objects be

$$Y_C(x, \sigma, \bar{e}; \mu) = \max\{0, x - \theta(\sigma, \bar{e}; \mu)\sqrt{x}\}, \quad (9)$$

$$W_C(x, \sigma, \bar{e}; \mu) = \max\left\{0, \beta + v(x) - \frac{a(\sigma, \bar{e})}{\theta(\sigma, \bar{e}; \mu)}x^2\right\}. \quad (10)$$

These formulas respect the basic continuation discipline of the model. First,

$$0 \leq Y_C(x, \sigma, \bar{e}; \mu) \leq x \quad \text{for every } x \in [0, \bar{x}],$$

so the residual burden under C never exceeds the burden under U . Second, $W_C \geq 0$ by construction, which is exactly the lapse convention encoded in (4). Third, both objects are continuous in x , Y_C is weakly decreasing in μ , and W_C is weakly increasing in μ on each untruncated branch. The court's strict-improvement rule (2) can therefore be evaluated directly from (9).

Witness specialization for Proposition 2. Appendix C uses

$$\tau(g, \bar{e}) = 10, \quad a(g, \bar{e}) = 5, \quad \tau(b, \bar{e}) = 3, \quad a(b, \bar{e}) = \frac{1}{100},$$

which yields

$$Y_C(x, g, \bar{e}; \mu) = \max\{0, x - 10\mu\sqrt{x}\}, \quad Y_C(x, b, \bar{e}; \mu) = \max\{0, x - 3\mu\sqrt{x}\},$$

$$W_C(x, g, \bar{e}; \mu) = \max\left\{0, 1 + \frac{x}{2} - \frac{x^2}{2\mu}\right\}, \quad W_C(x, b, \bar{e}; \mu) = \max\left\{0, 1 + \frac{x}{2} - \frac{x^2}{300\mu}\right\}.$$

For the three comparison values $\mu \in \{1/4, 11/20, 7/10\}$, the court uses the same top-level remedy map in both states. On the left of $h = 1$, C is feasible exactly when

$$\kappa + Y_C(x, \sigma, \bar{e}; \mu) < x \iff \kappa < \min\{x, \mu\tau(\sigma, \bar{e})\sqrt{x}\}.$$

Since $\kappa = 2/5$, C cannot beat U on $[0, 2/5]$. For $x > 2/5$, the smallest value of $\mu\tau(\sigma, \bar{e})\sqrt{x}$ over the comparison set occurs at $\mu = 1/4$, $\tau = 3$, and $x = 2/5$, giving

$$\frac{3}{4}\sqrt{\frac{2}{5}} > \frac{2}{5} = \kappa.$$

So on $(2/5, 1]$ the court chooses C in both states.

On the right of h , the court chooses C exactly when

$$\kappa + Y_C(x, \sigma, \bar{e}; \mu) < h \iff x - \mu\tau(\sigma, \bar{e})\sqrt{x} < 1 - \kappa = \frac{3}{5}.$$

The worst case is again $\mu = 1/4$ and $\tau = 3$. On $[1, 3/2]$ the function $x - \frac{3}{4}\sqrt{x}$ is increasing, and at the upper endpoint

$$\frac{3}{2} - \frac{3\sqrt{6}}{8} < \frac{3}{5}.$$

Hence C is chosen on $(1, 3/2]$ in both states as well. This verification is what allows Appendix C to compare the three μ values using a common remedy map.

Multiplicative witness class for endogenous design. For the investment example, let

$$\tau(\sigma, e) = \tau_0(\sigma)g_M(e), \quad a(\sigma, e) = a_0(\sigma),$$

where $g_M(0) = 1$, $g'_M(e) > 0$, and $g''_M(e) \leq 0$. Then

$$\theta(\sigma, e; \mu) = \mu\tau_0(\sigma)g_M(e),$$

so $\theta_e(\sigma, e; \mu) > 0$ and repairability design affects the model only through the continuation channel, as required by Assumption 2. Appendix F uses the concrete choice

$$g_M(e) = \sqrt{1 + e}.$$

That witness is sufficient to verify the local zero-margin implication in Theorem 2(b). It does not, by itself, imply any global comparative static in μ for the outer design problem.

Appendix B Probabilistic Review Extension

This appendix leaves the main text unchanged and varies only the assumption that review occurs with certainty. It does not introduce strategic litigants, filing costs, bargaining, or a replacement

benchmark theorem.

Fix an exogenous monotone review probability

$$p(x) = \frac{x}{\bar{x}}.$$

Let $V_B^{\text{review}}(x)$ and $V_E^{\text{review}}(x; \mu, \bar{e})$ denote the binary and extended values conditional on review under the main-text model. The ex ante objects become

$$V_B^P(x) = p(x)V_B^{\text{review}}(x) + [1 - p(x)](\beta + v(x))$$

and

$$V_E^P(x; \mu, \bar{e}) = p(x)V_E^{\text{review}}(x; \mu, \bar{e}) + [1 - p(x)](\beta + v(x)).$$

Benchmark modification. Under probabilistic review, the exact certain-review conclusion that the binary benchmark is pinned at h is no longer generic. In the baseline witness environment with $h = 1$, $\bar{x} = 3/2$, $\beta = 1$, and $v(x) = x/2$, however,

$$V_B^P(x) = 1 + \frac{x}{2} \quad (x \leq 1), \quad V_B^P(x) = 1 - \frac{x}{6} - \frac{x^2}{3} \quad (x > 1),$$

so the first branch is strictly increasing on $[0, 1]$ and the second is strictly decreasing on $(1, 3/2]$. Hence the baseline probabilistic-review benchmark still satisfies $x_B^{P,*} = 1$.

Suggestive numerical exploration. This appendix does not assert a theorem under probabilistic review. It records a suggestive numerical exploration using the same witness-environment primitives as Proposition 2. Under the same exogenous review schedule $p(x) = x/\bar{x}$ and the same continuation-side witness primitives, a deterministic μ -grid over $[0.1, 1.0]$ with step size 0.005 yields the optimizer path nonaggressive $\rightarrow$ aggressive $\rightarrow$ nonaggressive $\rightarrow$ aggressive, with approximate switch points near $\mu \in [0.175, 0.18]$, $\mu \in [0.23, 0.235]$, and $\mu \in [0.605, 0.61]$. These switch points were verified by bisection to four decimal places at each transition. At the original Proposition 2 comparison values, the extended-regime optimizer is nonaggressive at $\mu = 1/4$ and $\mu = 11/20$, but aggressive again at $\mu = 7/10$. The exact certain-review triple is therefore not preserved pointwise,

while non-monotonicity itself survives. Because this is a numerical exercise rather than an analytic result, the reported switches should be interpreted as evidence that non-monotonicity is qualitatively robust to probabilistic review rather than as precise predictions about switch-point locations.

Investment benchmark. When the review probability depends only on x , the fixed- x return to repairability investment is simply scaled by $p(x)$:

$$V_E^p(x; e_1) - V_E^p(x; e_0) = p(x)(V_E^{\text{review}}(x; e_1) - V_E^{\text{review}}(x; e_0)).$$

The appendix therefore leaves the main-text investment theorem benchmarked to the case $p = p(x)$ and makes no new global claim once x is allowed to readjust or once the review probability itself depends on e .

Appendix C Counterexample Calculations

This appendix records the exact branch calculations behind Proposition 2 together with a bounded local robustness check. The constructive counterexample itself is unchanged: it uses the three reference values $\mu = 1/4$, $\mu = 11/20$, and $\mu = 7/10$ and the same witness environment stated in the proposition. Figure 1 plots the exact piecewise value functions, and Figure 2 adds the (μ, κ) parameter map using the same witness primitives.

Table 1 should be read as a bounded local check rather than as a substitute for theorem-level comparative statics. The point is simply that the aggressive/nonaggressive/aggressive pocket around $\mu = 11/20$ persists on small neighborhoods of the witness environment and is therefore not an artifact of one exact primitive vector.

Remark 4 (Joint robustness). Table 1 varies one primitive at a time while holding others fixed. One-at-a-time robustness does not imply joint robustness: a simultaneous perturbation of $(\text{Pr}(g), \kappa, a(b))$ could in principle close the non-monotone pocket even if each univariate perturbation preserves it, because the three mechanisms (truncation kinks, left-envelope re-expansion, and attainment failure) could interact under joint perturbation in ways that cancel the non-monotonicity. However, the strict-inequality argument in the proof of Proposition 2 implies the following: since all value com-

Table 1: Local one-at-a-time robustness of the obstruction around the witness environment.

Primitive family	Tested neighborhood	Pattern survives?	Approximate transition windows in μ
$\text{Pr}(g)$	0.298 to 0.302	Yes	lower window [0.518, 0.530]; upper window [0.606, 0.615]
$\tau(g)$	9.8 to 10.2	Yes	lower window [0.513, 0.536]; upper window [0.598, 0.623]
$a(b)$	0.0096 to 0.0104	Yes	lower window [0.523, 0.525]; upper window [0.610, 0.611]
$\bar{x}$	1.490 to 1.510	Yes	lower window [0.515, 0.533]; upper window [0.610, 0.611]
κ	0.3950 to 0.4050	Yes	lower window [0.523, 0.524]; upper window [0.610, 0.611]

Notes: Each row perturbs one primitive around its baseline value while holding all others fixed. Baseline: $\text{Pr}(g) = 0.3$, $\tau(g) = 10$, $a(b) = 0.01$, $\bar{x} = 1.5$, $\kappa = 0.4$; held-fixed primitives: $h = 1$, $\beta = 1$, $v(x) = x/2$, $a(g) = 5$, $\tau(b) = 3$. Transition windows computed by bisection (four decimal places). Source: authors' calculations.

parisons at the three reference values of μ are strict, and both the value functions and the optimizer correspondence are continuous in the primitive vector, the set of primitive vectors that generate the non-monotone pattern is *open* in the product topology on the primitive space. An open set containing the witness vector necessarily contains an open ball around it, so joint perturbations within that ball preserve the pattern. Computing the exact radius of that ball (the joint robustness region) would require evaluating the value comparisons along the boundary of the non-monotone set, which is feasible numerically but beyond the scope of the current analysis. The key qualitative point is that openness guarantees the pattern is not fragile to joint perturbation at the witness vector itself. A sufficient condition for non-monotonicity in the general model is that the continuation value W_C exhibits state-dependent curvature that varies non-monotonically with μ , combined with truncation kinks at state-specific thresholds. The witness class verifies these conditions constructively; characterizing the full set of primitives that generate them is an open question.

Appendix D Local Boundary Remark

This appendix records the local boundary objects around $x = h$. They are useful diagnostics for the directional problem, but they do not replace the global comparisons in Proposition 2.

Fix $e = \bar{e}$ and write

$$\Delta(x; \mu) := V_E(x; \mu, \bar{e}) - V_B(x).$$

For a rightward deviation $z \in (0, \bar{x} - h]$, define

$$\Omega_+(z; \mu) := V_E(h + z; \mu, \bar{e}) - V_E(h; \mu, \bar{e}) = \Delta(h + z; \mu) - \Delta(h; \mu) - [\beta + v(h)].$$

For a leftward deviation $z \in (0, h]$, define

$$\Omega_-(z; \mu) := V_E(h - z; \mu, \bar{e}) - V_E(h; \mu, \bar{e}) = \Delta(h - z; \mu) - \Delta(h; \mu) - [v(h) - v(h - z)].$$

These are the exact one-sided level comparisons around the binary benchmark. The right object nets out the binary cliff $\beta + v(h)$; the left object nets out the ordinary policy-value loss from moving below h .

The benchmark boundary h is locally stable from the right if and only if there exists $\varepsilon > 0$ such that

$$\Omega_+(z; \mu) \leq 0 \quad \text{for all } z \in (0, \varepsilon].$$

It is locally stable from the left if and only if there exists $\varepsilon > 0$ such that

$$\Omega_-(z; \mu) \leq 0 \quad \text{for all } z \in (0, \varepsilon].$$

Equivalently, one may define the neighborhood diagnostics

$$\bar{\Omega}_+(\varepsilon; \mu) := \sup_{0 < z \leq \varepsilon} \Omega_+(z; \mu), \quad \bar{\Omega}_-(\varepsilon; \mu) := \sup_{0 < z \leq \varepsilon} \Omega_-(z; \mu),$$

and read local optimality of h as the joint condition $\bar{\Omega}_+(\varepsilon; \mu) \leq 0$ and $\bar{\Omega}_-(\varepsilon; \mu) \leq 0$ for some $\varepsilon > 0$.

This language is diagnostic rather than theorem-bearing. In the witness environment of Proposition 2, the decisive comparisons are not smooth first-order conditions at h . At $\mu = 11/20$, for example, the best aggressive value is a right-side supremum at the boundary, while the winning

nonaggressive policy is an interior point below h . That is exactly the sort of non-smooth geometry for which the level-comparison objects Ω_+ and Ω_- are informative, and exactly why I do not pursue a global monotone directional theorem.

Appendix E Proofs

Proof of Proposition 1. Under binary review the court compares only U and S . By (1), uphold yields judicial loss x and strike yields judicial loss h , with U chosen at $x = h$. The legislature's stage-1 value is therefore exactly (6):

$$V_B(x) = \begin{cases} \beta + v(x) & \text{if } x \leq h, \\ 0 & \text{if } x > h. \end{cases}$$

On $[0, h]$, V_B is strictly increasing because $v'(x) > 0$. Its unique maximizer on the survive region is therefore $x = h$. On $(h, \bar{x}]$, the value is identically zero. Assumption 1 imposes $\beta + v(h) > 0$, so

$$V_B(h) = \beta + v(h) > 0 = V_B(x) \quad \text{for all } x > h.$$

Hence the unique binary-review maximizer is h , and $X_B^* = \{h\}$. □

Proof of Theorem 1. Fix $e = \bar{e}$ and $\mu > 0$.

For $x \leq h$, the extended-regime value is

$$V_E(x; \mu, \bar{e}) = \beta + v(x) + \mathbb{E}_\sigma \left[(W_C(x, \sigma, \bar{e}; \mu) - \beta - v(x)) \mathbf{1}\{x \in R_C^U(\sigma, \bar{e}; \mu)\} \right].$$

For $x > h$, it is

$$V_E(x; \mu, \bar{e}) = \mathbb{E}_\sigma \left[W_C(x, \sigma, \bar{e}; \mu) \mathbf{1}\{x \in R_C^S(\sigma, \bar{e}; \mu)\} \right].$$

For part (a), suppose first that $x \leq h$ and that $x \notin R_C^U(\sigma, \bar{e}; \mu)$ in every state. Then the indicator in the first display is zero state by state, so

$$V_E(x; \mu, \bar{e}) = \beta + v(x) = V_B(x).$$

Now suppose that $x > h$ and that $x \notin R_C^S(\sigma, \bar{e}; \mu)$ in every state. Then the indicator in the second display is zero state by state, so

$$V_E(x; \mu, \bar{e}) = 0 = V_B(x).$$

Thus only statutes that fall in the conditional-remedy region on the relevant side of h can be repriced by the menu expansion.

For part (b), let $x > h$ satisfy

$$\Pr_\sigma[x \in R_{C,+}^S(\sigma, \bar{e}; \mu)] > 0.$$

Because $x > h$, binary review strikes and $V_B(x) = 0$. Under conditional review,

$$V_E(x; \mu, \bar{e}) = \mathbb{E}_\sigma[W_C(x, \sigma, \bar{e}; \mu) \mathbf{1}\{x \in R_C^S(\sigma, \bar{e}; \mu)\}].$$

On the event $\{x \in R_{C,+}^S\}$, the court chooses C and the continuation payoff is strictly positive by definition, so the random variable inside the expectation is strictly positive on an event of positive probability and nonnegative elsewhere. Therefore

$$V_E(x; \mu, \bar{e}) > 0 = V_B(x).$$

So some statute that is worthless under binary review acquires strictly positive value under conditional review. □

Proof of Proposition 2. Appendix C derives the exact piecewise formulas for the witness environment (7)–(8).

At $\mu = 1/4$, the value function has three branches and the rightmost branch is strictly increasing on its domain. Appendix C shows that

$$\max_{x \leq 2/5} V_E(x; 1/4, \bar{e}) = \frac{6}{5}, \quad \max_{2/5 < x \leq r_1} V_E(x; 1/4, \bar{e}) = \frac{4031}{3656},$$

while

$$V_E(3/2; 1/4, \bar{e}) = \frac{301}{250}.$$

Since $\frac{301}{250} > \frac{6}{5} > \frac{4031}{3656}$, the unique global maximizer is $x = 3/2$, so

$$X^*(1/4) = \left\{ \frac{3}{2} \right\}.$$

At $\mu = 11/20$, Appendix C shows that the nonaggressive middle branch has unique maximizer

$$x_L = \frac{825}{914} < 1$$

with value

$$V_E(x_L; 11/20, \bar{e}) = \frac{4481}{3656}.$$

On the aggressive side,

$$\sup_{1 < x \leq 3/2} V_E(x; 11/20, \bar{e}) = \frac{1009}{825},$$

and this right-side supremum is not attained on $(1, 3/2]$. Because

$$\frac{1009}{825} < \frac{4481}{3656},$$

the unique global optimizer is nonaggressive:

$$X^*(11/20) = \left\{ \frac{825}{914} \right\}.$$

At $\mu = 7/10$, both repaired-state quadratics are positive on the repaired region, so Appendix C yields the two-branch expression

$$V_E(x; 7/10, \bar{e}) = \begin{cases} 1 + \frac{x}{2}, & 0 \leq x \leq 2/5, \\ 1 + \frac{x}{2} - \frac{457}{2100}x^2, & 2/5 < x \leq 3/2. \end{cases}$$

The repaired-region quadratic has vertex at

$$x_R = \frac{525}{457} > 1$$

with value $\frac{2353}{1828}$. Appendix C also shows that

$$V_E(1; 7/10, \bar{e}) = \frac{2693}{2100} < \frac{2353}{1828}, \quad \max_{x \leq 2/5} V_E(x; 7/10, \bar{e}) = \frac{6}{5} < \frac{2353}{1828}.$$

Hence

$$X^*(7/10) = \left\{ \frac{525}{457} \right\}.$$

Combining the three cases gives

$$X^*(1/4) \cap (h, \bar{x}] \neq \emptyset, \quad X^*(11/20) \subset [0, h), \quad X^*(7/10) \cap (h, \bar{x}] \neq \emptyset.$$

Because $1/4 < 11/20 < 7/10$, no cutoff $\hat{\mu}$ can make aggressive optimality equivalent to either $\mu \geq \hat{\mu}$ or $\mu \leq \hat{\mu}$. The strict inequalities in the value comparisons imply that the same pattern survives on small neighborhoods of the three reference values. $\square$

Proof of Theorem 2. For part (a), Assumption 2 states that repairability design enters only through the contingent-remand channel. Binary review excludes C , so for every (x, e, μ) ,

$$V_B(x, e; \mu) = V_B(x, 0; \mu).$$

Therefore

$$\Psi_B(e; \mu) = -\phi(e) + \sup_{x \in [0, \bar{x}]} V_B(x, e; \mu) = -\phi(e) + \sup_{x \in [0, \bar{x}]} V_B(x, 0; \mu).$$

The second term is constant in e . Since design cost is increasing, $\phi(e) \geq \phi(0)$ for all $e \geq 0$, so

$$\Psi_B(e; \mu) \leq \Psi_B(0; \mu).$$

Hence $e = 0$ is optimal under binary review. If ϕ is strictly increasing, then $\phi(e) > \phi(0)$ for every $e > 0$, so the inequality is strict and $e = 0$ is the unique binary optimum.

For part (b), assume that $\Psi_E(\cdot; \mu)$ admits a right derivative at zero and that

$$\Psi'_{E,+}(0; \mu) > 0.$$

By the definition of the right derivative, there exists $\hat{e}(\mu) > 0$ such that

$$\frac{\Psi_E(e; \mu) - \Psi_E(0; \mu)}{e} > 0 \quad \text{for all } e \in (0, \hat{e}(\mu)].$$

Multiplying by $e > 0$ yields

$$\Psi_E(e; \mu) > \Psi_E(0; \mu) \quad \text{for all } e \in (0, \hat{e}(\mu)].$$

So $e = 0$ cannot be optimal under conditional review.

If the outer argmax is nonempty and some optimizer $\hat{e}$ satisfied $\hat{e} = 0$, then $\Psi_E(0; \mu)$ would equal the maximum of $\Psi_E(\cdot; \mu)$. The strict inequality just established would then be impossible. Therefore every optimizer must satisfy $e_E^*(\mu) > 0$. $\square$

Appendix F Multiplicative Repair-Technology Example

This appendix records one constructive witness-class example for Theorem 2(b). The example is illustrative only. It does not add a new theorem, prove uniqueness of optimal repairability design, or recover any monotone comparative static in μ .

Take the same baseline witness environment as Proposition 2 and define

$$g_M(e) = \sqrt{1 + e}, \quad \tau(\sigma, e) = \tau_0(\sigma)g_M(e), \quad a(\sigma, e) = a_0(\sigma), \quad \phi(e) = 10^{-3}e,$$

with

$$\mu = \frac{\sqrt{10}}{3}, \quad \tau_0(g) = 10, \quad \tau_0(b) = 3, \quad a_0(g) = 5, \quad a_0(b) = \frac{1}{100}.$$

The value $\mu = \sqrt{10}/3 \approx 1.054$ is chosen to place the environment in the aggressive region of the (μ, κ) phase diagram (Figure 2), where the conditional remedy is actively used on the strike side and the investment margin is operative. At this μ , both repaired-state quadratics are positive on the repaired region, so the conditional remedy creates positive continuation value on the aggressive side, exactly the configuration needed to verify that the marginal benefit of repairability design at $e = 0$ is strictly positive.

Under binary review, repairability design is dormant, so

$$\Psi_B(e; \mu) = \frac{3}{2} - 10^{-3}e.$$

The binary outer objective is therefore strictly decreasing and uniquely maximized at $e = 0$.

For the conditional-review witness, write $d = x - 1$ on the aggressive side. The change of variable $d := x - 1$ centers the quadratic at the strike threshold; this is algebraically equivalent to a change of variable that shifts the quadratic center; the functional form remains within the same witness class. The resulting repaired continuation values are

$$W_C(x, g, e) = 1 + \frac{x}{2} - \frac{9}{2500} \frac{d^2}{1+e}, \quad W_C(x, b, e) = 1 + \frac{x}{2} - \frac{1}{25} \frac{d^2}{1+e}.$$

On the branch where both states strictly open the conditional remedy, this implies

$$V_E(x, e; \mu) = 1 + \frac{x}{2} - \frac{727}{25000} \frac{(x-1)^2}{1+e}.$$

That branch is strictly increasing on $x \in (1, 3/2]$, so

$$\sup_{x \in [0, 3/2]} V_E(x, e; \mu) = \frac{7}{4} - \frac{727}{100000(1+e)}.$$

Hence

$$\Psi_E(e; \mu) = \frac{7}{4} - \frac{727}{100000(1+e)} - 10^{-3}e.$$

At zero,

$$\Psi_E(0; \mu) = 1.74273, \quad \Psi'_{E,+}(0; \mu) = \frac{727}{100000} - 10^{-3} = 0.00627 > 0.$$

So the local zero-margin implication is active: the conditional-review outer objective rises immediately to the right of zero, even though the binary benchmark continues to select $e = 0$. In this calibration the witness curve peaks near

$$e^* \approx \sqrt{7.27} - 1 \approx 1.696.$$

At $e = 0$ the aggressive-side supremum is approached from $x \uparrow 3/2$ rather than attained exactly at $x = 3/2$, because the bad state is exactly tied there under the strict-improvement rule.

Appendix G Welfare Note for the Witness Economy

This appendix records the ex ante payoff comparison between conditional and binary review at the three reference values of μ used in Proposition 2. The comparison is specific to the maintained witness economy and does not constitute a general welfare result.

Under binary review, the legislature chooses $x_B^* = h = 1$, earning $V_B(h) = \beta + v(h) = 3/2$. The court's ex ante loss is $L_J^B = h = 1$. Under conditional review at each reference μ , the legislature's ex ante payoff $V_E(x_E^*; \mu, \bar{e})$ and the court's ex ante loss $L_J^E(\mu)$ can be computed from the witness-class formulas in Appendix A and the counterexample calculations in Appendix C. At all three reference values ($\mu = 1/4$, $\mu = 11/20$, and $\mu = 7/10$) the court's ex ante loss under conditional review is strictly below the binary benchmark ($L_J^E < 1$), while the legislature's ex ante payoff under conditional review is strictly below the binary benchmark ($V_E < 3/2$). The conditional remedy therefore benefits the court and harms the legislature in expectation at these parameter values. Neither regime ex ante Pareto-dominates the other, and aggregate welfare cannot be signed without specifying an explicit social welfare function or aggregator that weights the two actors' objectives.

This conflict reinforces the main text's welfare non-claim (Section 9): signing the welfare effect of conditional review requires normative structure beyond what the positive model supplies.

References

- Bird, Brian. 2019. "The Judicial Notwithstanding Clause: Suspended Declarations of Invalidity," *Manitoba Law Journal* 42(1): 1–30.
- Bueno de Mesquita, Ethan, and Matthew C. Stephenson. 2002. "Informative Precedent and Intrajudicial Communication," *American Political Science Review* 96(4): 755–766.
- Butt, Simon. 2015. *The Constitutional Court and Democracy in Indonesia*. Leiden: Brill.
- Butt, Simon, and Prayekti Murharjanti. 2022. "What Constitutes Compliance? Legislative Responses to Constitutional Court Decisions in Indonesia," *International Journal of Constitutional Law* 20(1): 428–453.
- Cameron, Charles M. 2000. *Veto Bargaining: Presidents and the Politics of Negative Power*. Cambridge: Cambridge University Press.
- Dixon, Rosalind. 2007. "Creating Dialogue about Socioeconomic Rights: Strong-Form Versus Weak-Form Judicial Review Revisited," *International Journal of Constitutional Law* 5(3): 391–418.
- Dixon, Rosalind, and Tom Ginsburg. 2017. "The Forms and Limits of Constitutions as Political Insurance," *International Journal of Constitutional Law* 15(4): 988–1012.
- Fox, Justin, and Matthew C. Stephenson. 2011. "Judicial Review as a Response to Political Posturing," *American Political Science Review* 105(2): 397–414.
- Ginsburg, Tom, and Mila Versteeg. 2014. "Why Do Countries Adopt Constitutional Review?" *Journal of Law, Economics, and Organization* 30(3): 587–622.
- Fish, Eric S. 2016. "Choosing Constitutional Remedies," *UCLA Law Review* 63: 322–395.
- Hendrianto, Stefanus. 2018. *Law and Politics of Constitutional Courts: Indonesia and the Search for Judicial Heroes*. London: Routledge.
- Hogg, Peter W., and Allison A. Bushell. 1997. "The Charter Dialogue Between Courts and Legislatures (Or Perhaps the Charter of Rights Isn't Such a Bad Thing After All)," *Osgoode Hall Law Journal* 35(1): 75–124.

- Hogg, Peter W., Allison A. Bushell Thornton, and Wade K. Wright. 2007. "Charter Dialogue Revisited—Or 'Much Ado About Metaphors,'" *Osgoode Hall Law Journal* 45(1): 1–65.
- Kavanagh, Aileen. 2015. "What's So Weak about 'Weak-Form Review'? The Case of the UK Human Rights Act 1998," *International Journal of Constitutional Law* 13(4): 1008–1039.
- Klug, Heinz. 2000. *Constituting Democracy: Law, Globalism and South Africa's Political Reconstruction*. Cambridge: Cambridge University Press.
- Kommers, Donald P., and Russell A. Miller. 2012. *The Constitutional Jurisprudence of the Federal Republic of Germany*. 3rd ed. Durham: Duke University Press.
- Krehbiel, Jay N. 2021. "Strategic Delay and the Use of Incompatibility Rulings at the German Constitutional Court," *Journal of Politics* 83(3): 1071–1084.
- Lax, Jeffrey R. 2011. "The New Judicial Politics of Legal Doctrine," *Annual Review of Political Science* 14: 131–157.
- Leckey, Richard. 2016. "The Harms of Remedial Discretion," *International Journal of Constitutional Law* 14(3): 584–607.
- Shipan, Charles R. 2004. "Regulatory Regimes, Agency Actions, and the Conditional Nature of Congressional Influence," *American Political Science Review* 98(3): 467–480.
- Stephenson, Matthew C. 2004. "Court of Public Opinion: Government Accountability and Judicial Independence," *Journal of Law, Economics, and Organization* 20(2): 379–399.
- Roux, Theunis. 2013. *The Politics of Principle: The First South African Constitutional Court, 1995–2005*. Cambridge: Cambridge University Press.
- Roach, Kent. 2004. "Dialogic Judicial Review and its Critics," *Supreme Court Law Review* 23: 49–104.
- Rogers, James R., and Georg Vanberg. 2002. "Judicial Advisory Opinions and Legislative Outcomes in Comparative Perspective," *American Journal of Political Science* 46(2): 379–397.
- Rogers, James R., and Georg Vanberg. 2007. "Resurrecting Lochner: A Defense of Unprincipled Judicial Activism," *Journal of Law, Economics, and Organization* 23(2): 442–468.
- Schlaich, Klaus, and Stefan Koriath. 2004. *Das Bundesverfassungsgericht: Stellung, Verfahren,*

Entscheidungen. 7th ed. Munich: C.H. Beck.

Turner, Ian R. 2017. “Working Smart and Hard? Agency Effort, Judicial Review, and Policy Precision,” *Journal of Theoretical Politics* 29(1): 69–96.

Vanberg, Georg. 2005. *The Politics of Constitutional Review in Germany*. Cambridge: Cambridge University Press.

Vanberg, Georg. 1998. “Abstract Judicial Review, Legislative Bargaining, and Policy Compromise,” *Journal of Theoretical Politics* 10(3): 299–326.

Wicaksono, Dian Agung, and Faiz Rahman. 2022. “Influencing or Intervention? Impact of Constitutional Court Decisions on the Supreme Court in Indonesia,” *Constitutional Review* 8(2): 260–294.